



ISTN7MP MOBILE AND PERSONAL INFORMATION SYSTEMS HONORS MODULE – SEMESTER 1 - 2026

PROF PRABHAKAR RONTALA SUBRAMANIAM

WEEK 1: MODULE INTRODUCTION

- BY PROF. IRENE GOVENDER

WEEK 2 – INTRODUCTION TO JAVA

- BY PROF. IRENE GOVENDER

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WEEK 3 – JAVA & XML (CONTD...)

- BY PROF. IRENE GOVENDER

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MOBILE AND PERSONAL INFORMATION SYSTEMS

- PART I : MOBILE COMPUTING
- PART II : MOBILE AND PERSONAL INFORMATION SYSTEMS
- PART III : ANDROID MOBILE APP DEVELOPMENT

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WEEK 4 – MOBILE COMPUTING

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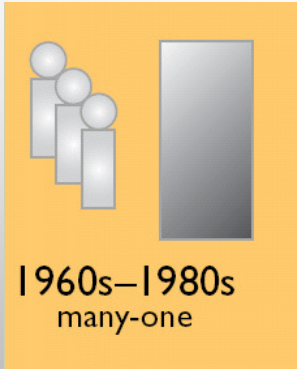
PART I

MOBILE COMPUTING

MOBILE COMPUTING TECHNOLOGIES

WEEK 4

COMPUTING EVOLUTION

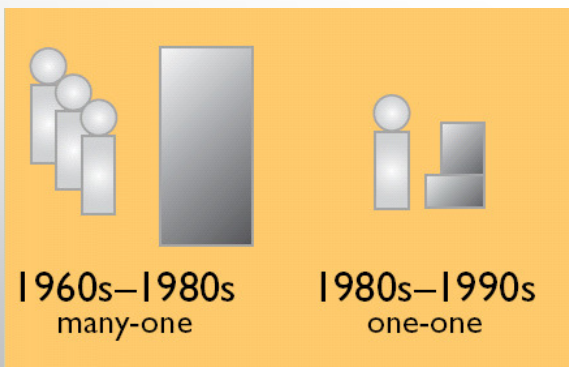


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COMPUTING EVOLUTION

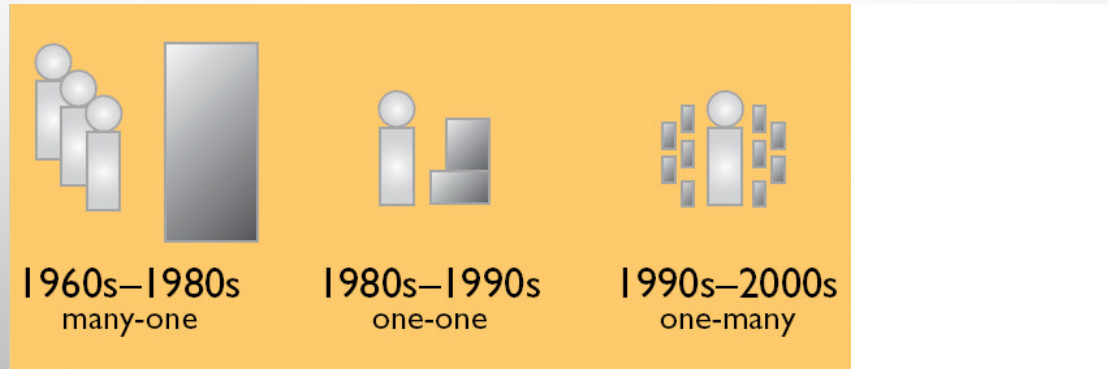


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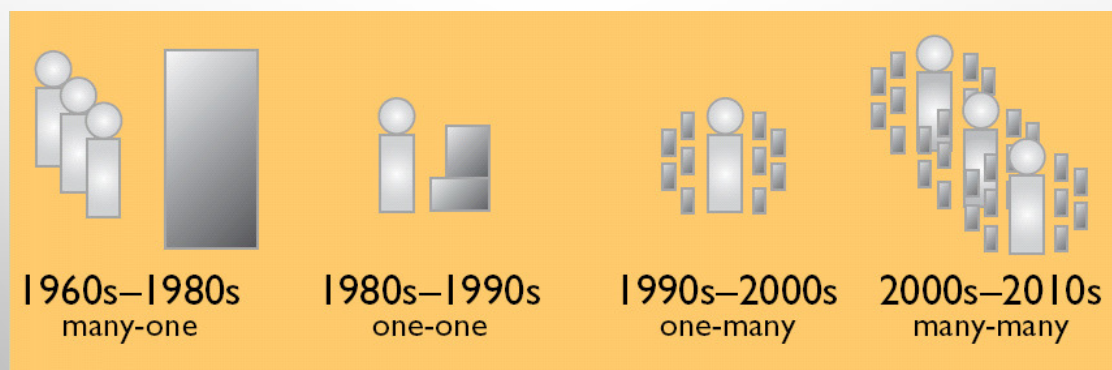


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COMPUTING EVOLUTION

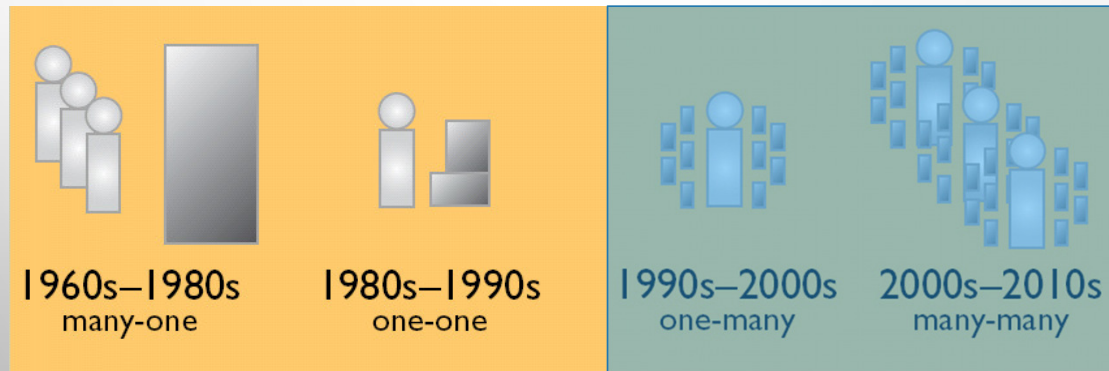


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COMPUTING EVOLUTION



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INTRODUCTION

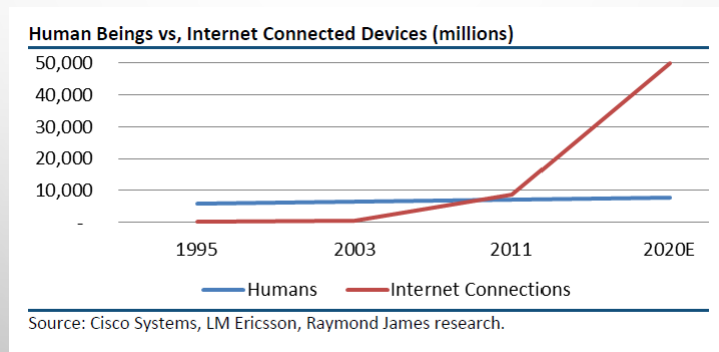
- DEVICE SIZE REDUCTION – SMALLER AND SMALLER
- PRICE REDUCTION
- OPTIMISED ENERGY CONSUMPTION
- PARADIGM SHIFT FROM H2M INTERACTION TO M2M
- INCREASED INTERNET USAGE AND BANDWIDTH
- INCREASED INTEGRATION OF COMMUNICATION TECHNOLOGY IN EVERYDAY COMPONENTS/OBJECTS
- INCREASING AWARENESS
- AS A RESULT, DEVICES ARE GETTING SMARTER..... AND SMARTER

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PAST.....PRESENT.....AND FUTURE



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INTERNET OF THINGS - INTRODUCTION

- THE "INTERNET OF THINGS" (IOT) IS BECOMING AN INCREASINGLY GROWING TOPIC OF CONVERSATION BOTH IN THE WORKPLACE AND OUTSIDE OF IT.
- IT'S A CONCEPT THAT NOT ONLY HAS THE POTENTIAL TO IMPACT HOW WE LIVE BUT ALSO HOW WE WORK. BUT WHAT EXACTLY IS THE "INTERNET OF THINGS"?
- THIS INCLUDES EVERYTHING FROM CELLPHONES, COFFEE MAKERS, WASHING MACHINES, HEADPHONES, LAMPS, WEARABLE DEVICES AND ALMOST ANYTHING ELSE YOU CAN THINK OF.
- THE NEW RULE FOR THE FUTURE IS GOING TO BE, "ANYTHING THAT CAN BE CONNECTED, WILL BE CONNECTED."

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INTERNET OF THINGS – INTRODUCTION (CONTD...)

- THE **IOT** IS THE INTERCONNECTION OF UNIQUELY IDENTIFIABLE EMBEDDED COMPUTING (PHYSICAL) DEVICES WITHIN THE EXISTING INTERNET INFRASTRUCTURE.
- DEVICES CAN BE EMBEDDED WITH ELECTRONICS, SOFTWARE, SENSORS, ACTUATORS, AND NETWORK CONNECTIVITY THAT ENABLE THESE DEVICES TO SHARE DATA.
- TYPICALLY, IOT IS EXPECTED TO OFFER ADVANCED CONNECTIVITY OF DEVICES, SYSTEMS, AND SERVICES THAT GOES BEYOND MACHINE-TO-MACHINE COMMUNICATIONS (M2M) AND COVERS A VARIETY OF PROTOCOLS, DOMAINS, AND APPLICATIONS.

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INTERNET OF THINGS – INTRODUCTION (CONTD...)

- GLOBAL STANDARDS INITIATIVE ON INTERNET OF THINGS (IOT-GSI) DEFINES IOT AS “THE INFRASTRUCTURE OF THE INFORMATION SOCIETY”.
- IOT IS NOT A SINGLE TECHNOLOGY, IT’S A CONCEPT IN WHICH MOST NEW THINGS ARE CONNECTED AND ENABLED.
- THE GOAL OF IOT IS TO ENABLE ANYTHING TO BE CONNECTED WITH EACH OTHER, ANYTIME, ANYPLACE AND ANYONE OPTIMALLY USING ANY NETWORK AND ANY SERVICE.

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PERVASIVE COMPUTING - INTRODUCTION

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PERVASIVE COMPUTING - INTRODUCTION

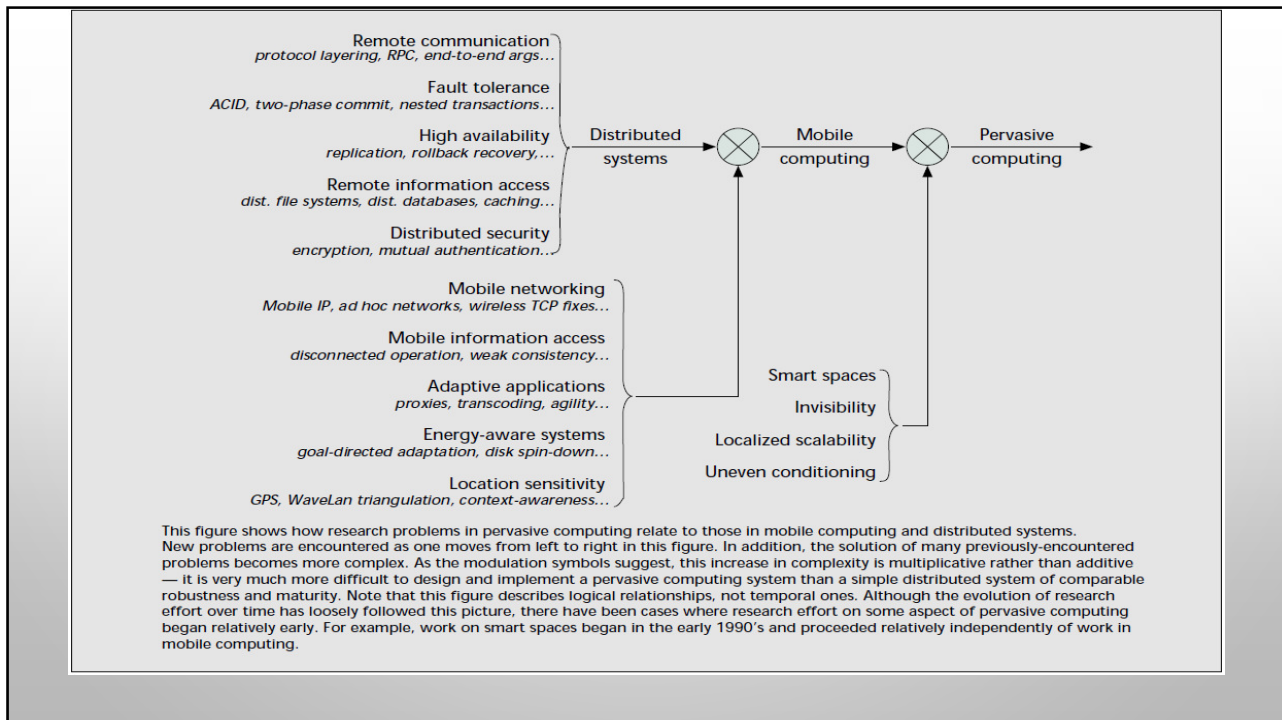
- PERVASIVE COMPUTING IS AN EMERGING TREND ASSOCIATED WITH EMBEDDING MICROPROCESSORS IN DAY-TO-DAY OBJECTS, ALLOWING THEM TO COMMUNICATE INFORMATION.
- IT IS ALSO KNOWN AS UBIQUITOUS COMPUTING. THE TERMS UBIQUITOUS AND PERVASIVE SIGNIFY "EXISTING EVERYWHERE." PERVASIVE COMPUTING SYSTEMS ARE TOTALLY CONNECTED AND CONSISTENTLY AVAILABLE.
- PROVIDES DECENTRALIZED (DISTRIBUTED) COMPUTATIONS ON DIVERSIFIED DEVICES AND NETWORKS WITH DIFFERENT PROTOCOLS VIA MOBILE COMMUNICATION STANDARDS.

(SOURCE: WWW.TECHOPEDIA.COM)

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MOBILE COMPUTING → PERVASIVE COMPUTING

MOBILE COMPUTING ← PERVASIVE COMPUTING

- MOBILE COMPUTING IS A BROAD GENERIC TERM USED TO DESCRIBE THE ABILITY OF THE SYSTEM TO CONNECT WIRELESSLY, USE CENTRALLY LOCATED DATA THROUGH SOFTWARE APPLICATIONS USING MOBILE COMMUNICATION AND PROCESSING DEVICES.
- PROVIDES DECENTRALIZED (DISTRIBUTED) COMPUTATIONS ON DIVERSIFIED DEVICES AND NETWORKS WITH DIFFERENT PROTOCOLS VIA MOBILE COMMUNICATION STANDARDS.

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MOBILE COMPUTING – AN INTRODUCTION

- MOBILE COMPUTING IS HUMAN-COMPUTER INTERACTION BY WHICH A COMPUTER IS EXPECTED TO BE TRANSPORTED DURING NORMAL USAGE, WHICH ALLOWS FOR TRANSMISSION OF DATA, VOICE AND VIDEO.
- MOBILE COMPUTING INVOLVES MOBILE COMMUNICATION, MOBILE HARDWARE, AND MOBILE SOFTWARE.
 - COMMUNICATION ISSUES INCLUDE NETWORKS WITH OR WITHOUT INFRASTRUCTURE AS WELL AS COMMUNICATION PROPERTIES, PROTOCOLS, DATA FORMATS AND CONCRETE TECHNOLOGIES.
 - HARDWARE INCLUDES MOBILE DEVICES OR DEVICE COMPONENTS.
 - MOBILE SOFTWARE DEALS WITH THE CHARACTERISTICS AND REQUIREMENTS OF MOBILE APPLICATIONS.

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(SOURCE: WIKIPEDIA)

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MOBILE COMPUTING - COMMUNICATION

- CELLULAR NETWORKS – NETWORKS WITH INFRASTRUCTURE
 - 0G NETWORK
 - 1G NETWORK
 - 2G NETWORK
 - 3G NETWORK
 - 4G NETWORK
 - 5G NETWORK
- WIRELESS AD HOC NETWORKS – NETWORK WITHOUT INFRASTRUCTURE
 - MOBILE AD HOC NETWORKS
 - SENSOR NETWORKS
- VEHICULAR AD HOC NETWORKS – HYBRID NETWORK
- SOFTWARE DEFINED NETWORKS (SDN)

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WIRELESS AD HOC NETWORKS

- TEMPORARY NETWORKS – ESTABLISHED FOR A PURPOSE
- DO NOT RELY ON INFRASTRUCTURE SUCH AS ROUTERS OR SWITCHES
- SELF ORGANIZING AND ADAPTIVE NETWORK
- SUPPORTS PEER-TO-PEER AND PEER-TO-REMOTE COMMUNICATIONS
- NETWORK FORMED WITHOUT ANY CENTRAL ADMINISTRATION
- MOBILE DEVICES ARE THE NODES IN THE NETWORK

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AD HOC NETWORKS

- NODES ARE COMPUTING DEVICES
- NODES CONNECTED USING WIRELESS COMMUNICATION TECHNOLOGIES
- ASSUME RANGE OF COMMUNICATION IS 5 METERS

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AD HOC NETWORKS

- NODES ARE COMPUTING DEVICES
- NODES CONNECTED USING COMMUNICATION TECHNOLOGIES
- ASSUME RANGE OF COMMUNICATION IS 5 METERS
- THESE DEVICES AT A DISTANCE OF 8 METERS CANNOT COMMUNICATE WITH EACH OTHER.
- NODES ARE OUT OF RANGE



8 meters



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AD HOC NETWORKS

- NODES ARE COMPUTING DEVICES
- NODES CONNECTED USING COMMUNICATION TECHNOLOGIES
- ASSUME RANGE OF COMMUNICATION IS 5 METERS
- PRESENCE OF INTERMEDIATE NODE BRINGS DEVICES WITHIN THE COMMUNICATION RANGE
- NOW COMMUNICATION IS POSSIBLE



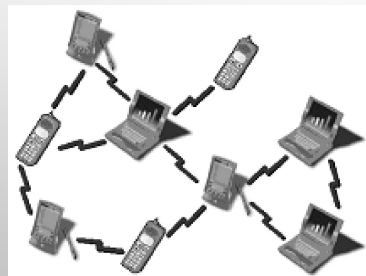
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MOBILE AD HOC NETWORKS (MANET)

- WHEN NODES ARE ON THE FLY (MOBILE) THEY ARE CALLED MANETS



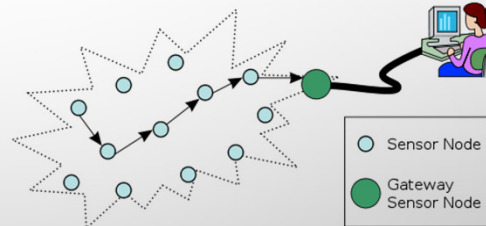
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SENSOR NETWORKS

- WIRELESS SENSOR NETWORKS (WSN) CAN ALSO BE CALLED AS WIRELESS SENSOR AND ACTUATOR NETWORKS (WSAN)
- SPATIALLY DISTRIBUTED WIRELESS SENSORS CAN MONITOR PHYSICAL ENVIRONMENTAL CONDITIONS SUCH AS TEMPERATURE, SOUND, PRESSURE, ETC.



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SENSOR NETWORKS - APPLICATIONS

- AREA MONITORING BY MILITARY FOR ENEMY INTRUSIONS
- HEALTH CARE MONITORING IN HOSPITALS BY WEARABLES AND IMPLANTED DEVICES
- AIR POLLUTION MONITORING
- FOREST FIRE DETECTION
- WATER QUALITY MONITORING
- AND MANY MORE

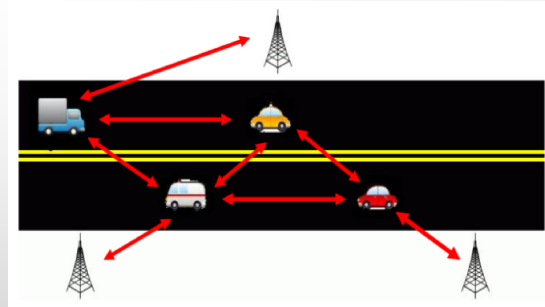
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VEHICULAR AD HOC NETWORKS (VANET)

- ASSUME VEHICLES IN PLACE OF MOBILE NODES
- THIS LEADS TO INTELLIGENT TRANSPORTATION SYSTEM
- VANETS ARE OUTGROWTH OF MANETS
- CAN COMMUNICATE TO OTHER VEHICLES AND ROAD SIDE INFRASTRUCTURE



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GENERAL CHARACTERISTICS

MANET

- DYNAMIC TOPOLOGY
- BANDWIDTH AND ENERGY CONSTRAINT
- INFRASTRUCTURELESS
- UNCONTROLLED MOVING PATTERNS
- LIMITED STORAGE AND PROCESSING CAPABILITY

VANET

- DYNAMIC TOPOLOGY
- ENERGY CONSTRAINT
- INFRASTRUCTURELESS TO SOME EXTENT
- UNCONTROLLED MOVING PATTERNS
- LIMITED STORAGE AND PROCESSING CAPABILITY
- HIGH NODE DENSITY
- HIGH MOBILITY SPEED
- MOBILITY ON CONTROLLED PATHS
- DRIVER BEHAVIOUR

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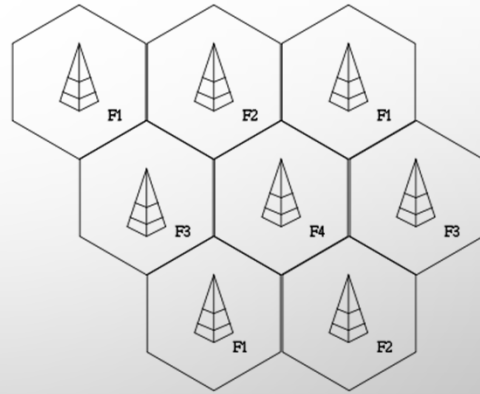
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CELLULAR NETWORK

- A RADIO SERVICE COMMUNICATION NETWORK IN WHICH LAST LINK IS WIRELESS
- LAND DISTRIBUTED NETWORK AREAS CALLED CELLS
- CELLS ARE REGULAR SHAPED – HEXAGONAL, SQUARE, CIRCULAR OR ANY REGULAR SHAPE

(SOURCE: WIKIPEDIA)



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CELLULAR TECHNOLOGIES

- GSM – GLOBAL SYSTEM FOR MOBILE COMMUNICATIONS
 - TDMA – TIME DIVISION MULTIPLE ACCESS
 - FDMA – FREQUENCY DIVISION MULTIPLE ACCESS
- CDMA - CODE DIVISION MULTIPLE ACCESS
- LTE (LONG TERM EVOLUTION)

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CELLULAR TECHNOLOGIES

- GSM – GLOBAL SYSTEM FOR MOBILE COMMUNICATIONS

- TDMA – TIME DIVISION MULTIPLE ACCESS

IN TDMA AND DIGITAL TECHNOLOGY, THE 12.5 KHZ CHANNEL BANDWIDTH IS MAINTAINED. THE RF SPECTRUM EFFICIENCY IS ACHIEVED WHEN TWO VOICE CHANNELS SHARE TIME TO CREATE A 6.25 KHZ EQUIVALENCY. WHICH MEANS, TDMA ALLOWS SEVERAL USERS TO SHARE THE SAME FREQUENCY CHANNEL BY DIVIDING THE SIGNAL INTO DIFFERENT TIME SLOTS. TDMA TECHNOLOGY IS POSSIBLE ONLY BY USING INTELLIGENT INFRASTRUCTURE TO MAKE AND CONTROL THE TIME SLOTS.

- FDMA – FREQUENCY DIVISION MULTIPLE ACCESS

- CDMA - CODE DIVISION MULTIPLE ACCESS

- LTE (LONG TERM EVOLUTION)

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CELLULAR TECHNOLOGIES

- GSM – GLOBAL SYSTEM FOR MOBILE COMMUNICATIONS

- TDMA – TIME DIVISION MULTIPLE ACCESS

- FDMA – FREQUENCY DIVISION MULTIPLE ACCESS - FDMA SPLITS THE TOTAL BANDWIDTH INTO MULTIPLE CHANNELS.

FREQUENCY-DIVISION MULTIPLE ACCESS (FDMA) IS A CHANNEL ACCESS METHOD USED IN SOME MULTIPLE-ACCESS PROTOCOLS. FDMA ALLOWS MULTIPLE USERS TO SEND DATA THROUGH A SINGLE COMMUNICATION CHANNEL, SUCH AS A COAXIAL CABLE OR MICROWAVE BEAM, BY DIVIDING THE BANDWIDTH OF THE CHANNEL INTO SEPARATE NON-OVERLAPPING FREQUENCY SUB-CHANNELS AND ALLOCATING EACH SUB-CHANNEL TO A SEPARATE USER. USERS CAN SEND DATA THROUGH A SUB-CHANNEL BY MODULATING IT ON A CARRIER WAVE AT THE SUB-CHANNEL'S FREQUENCY. FDMA SPLITS THE TOTAL BANDWIDTH INTO MULTIPLE CHANNELS.

- CDMA - CODE DIVISION MULTIPLE ACCESS

- LTE (LONG TERM EVOLUTION)

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CELLULAR TECHNOLOGIES (CONTD...)

- GSM – USES A SIM CARD
 - CHANNEL ACCESS METHOD FOR SHARED MEDIUM NETWORK
 - ALLOWS USERS TO SHARE SAME FREQUENCY CHANNEL
 - ALLOWS MULTIPLE STATIONS TO SHARE THE SAME TRANSMISSION MEDIUM
 - SIGNALS IN THE MEDIUM ARE DIVIDED INTO MULTIPLE TIME SLOTS
 - MOBILE STATIONS LISTEN AND BROADCAST ONLY IN ASSIGNED TIME SLOT
 - DURING THE REMAINING TIME PERIOD, THE MOBILE STATION DETECT SURROUNDING TRANSMITTERS IN DIFFERENT FREQUENCIES
 - SEVERAL TRANSMITTERS CONNECTED TO ONE RECEIVER (MULTIPLEXING)

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CELLULAR TECHNOLOGIES (CONTD...)

- CDMA – DOES NOT HAVE A SIM SLOT

CODE-DIVISION MULTIPLE ACCESS (CDMA) IS A CHANNEL ACCESS METHOD USED BY VARIOUS RADIO COMMUNICATION TECHNOLOGIES. CDMA IS AN EXAMPLE OF MULTIPLE ACCESS, WHERE SEVERAL TRANSMITTERS CAN SEND INFORMATION SIMULTANEOUSLY OVER A SINGLE COMMUNICATION CHANNEL. THIS ALLOWS SEVERAL USERS TO SHARE A BAND OF FREQUENCIES.

- CHANNEL ACCESS METHOD FOR SHARED MEDIUM NETWORK
- ALLOWS USERS TO SHARE SAME FREQUENCY CHANNEL
- ALLOWS MULTIPLE STATIONS TO SHARE THE SAME TRANSMISSION MEDIUM
- ENCODES INDIVIDUAL CONVERSATION INTO PSEUDO-RANDOMIZED DIGITAL SEQUENCE
- VOICE DATA REMAINS PROTECTED WHERE ONLY PARTICIPANTS IN THE PHONE CALL RECEIVES THE DATA

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CELLULAR TECHNOLOGIES (CONTD...)

- LTE – LONG TERM EVOLUTION
 - DISCUSSED AFTER CELLULAR NETWORK GENERATIONS

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CELLULAR NETWORK GENERATIONS

- 1 G (FIRST GENERATION)
 - EVOLUTION – EARLY 1980S (1970 – 1984)
 - ANALOG TECHNOLOGY
 - DATA BANDWIDTH – 2 Kbps
 - CAN MAKE PHONE CALLS IN ONE COUNTRY
 - APPLICATIONS – PAGING, CORDLESS TELEPHONES, PRIVATE MOBILE RADIO
- 2 G (SECOND GENERATION)
 - EVOLUTION – LATE 1980S (1980 – 1999)
 - ANALOG REPLACED WITH DIGITAL TECHNOLOGY – GSM (WITH TDMA) & CDMA
 - TIME DIVISION MULTIPLE ACCESS & CODE DIVISION MULTIPLE ACCESS
 - ENHANCED SPECTRUM EFFICIENCY – 14.4 – 64 Kbps
 - BETTER ENHANCED FEATURES LIKE ROAMING WAS INTRODUCED APART FROM VOICE, SMS, ETC.

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CELLULAR NETWORK GENERATIONS (CONTD...)

- 3 G (THIRD GENERATION)
 - EVOLUTION – 1990 - 2002
 - BANDWIDTH – 2 MBPS
 - INCREASED SPEED OF COMMUNICATION
 - SIMULTANEOUS USE OF VOICE AND DATA
 - APPLICATIONS – IN ADDITION, VIDEO CALLING, LIVE STREAMING, MOBILE INTERNET ACCESS, IPTV, ETC. BECAME POSSIBLE

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CELLULAR NETWORK GENERATIONS (CONTD...)

- 4 G (FOURTH GENERATION)
 - EVOLUTION – 2000 – PRESENT
 - BANDWIDTH SPECTRUM – 200 MBPS – 1 GBPS
 - LONG TERM EVOLUTION (LTE) TECHNOLOGY
 - REFERRED TO AS MAGIC (MOBILE MULTIMEDIA, ANY-WHERE, GLOBAL MOBILITY SOLUTIONS OVER, INTEGRATED WIRELESS AND CUSTOMISED SERVICES)
 - APPLICATIONS – IN ADDITION, IP TELEPHONY, ULTRA-BROADBAND INTERNET ACCESS, GAMING SERVICES, HIGH DEFINITION TV (HDTV) STREAMED MULTIMEDIA.

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CELLULAR NETWORK GENERATIONS (CONTD...)

- 5 G (FIFTH GENERATION)
 - EVOLUTION – EVOLVING
 - BANDWIDTH – 1 GBPS & HIGHER
 - LTE ADVANCED TECHNOLOGY (SPECTRUM EFFICIENT TECHNOLOGY)
 - ADDRESSES NETWORK CONGESTION, ENERGY CONSUMPTION, COST, RELIABILITY, ETC.
 - APPLICATIONS – USE OF HETEROGENEOUS NETWORK WITH 3 TO 5 TIMES BOOST IN THE NETWORK CAPACITY, BETTER SIGNAL RANGE, ETC.
- 2 G VS 3 G
 - WHILE PHONE AT IDLE BATTERY CONSUMPTION IS SIMILAR
 - 3 G DRAINS MORE BATTERY POWER WHILE TRANSFERRING DATA
 - 2 G DRAINS POWER CONSISTENTLY WHILE 3 G DRAINS WITH FLUCTUATIONS

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CELLULAR TECHNOLOGIES (CONTD...)

- LTE – LONG TERM EVOLUTION
 - FACILITIES OF BOTH GSM AND CDMA
 - FROM 4G
 - LETS WATCH THIS VIDEO AND UNDERSTAND BETTER
- 5G – ON ITS WAY
 - [HTTPS://WWW.YOUTUBE.COM/WATCH?V=GEX_D0SJVS0](https://www.youtube.com/watch?v=GEX_D0SJVS0)



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SOFTWARE DEFINED NETWORKS (SDN)

(SOURCE: [HTTPS://WWW.OPENNWORKING.ORG/SDN-RESOURCES/SDN-DEFINITION](https://www.opennetworking.org/sdn-resources/sdn-definition))

- AN ARCHITECTURE THAT IS DYNAMIC, MANAGEABLE, COST-EFFECTIVE AND ADAPTABLE
- IDEAL FOR HIGH-BANDWIDTH AND DYNAMIC NATURE OF TODAY'S APPLICATIONS
- DECOUPLES THE NETWORK CONTROL AND FORWARDING FUNCTIONS
- NETWORK CONTROL IS DIRECTLY PROGRAMMABLE
- UNDERLYING INFRASTRUCTURE IS ABSTRACTED FOR APPLICATIONS AND NETWORK SERVICES
- OPENFLOW PROTOCOL IS A FOUNDATIONAL ELEMENT FOR BUILDING SDN SOLUTIONS

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SDN ARCHITECTURE FEATURES

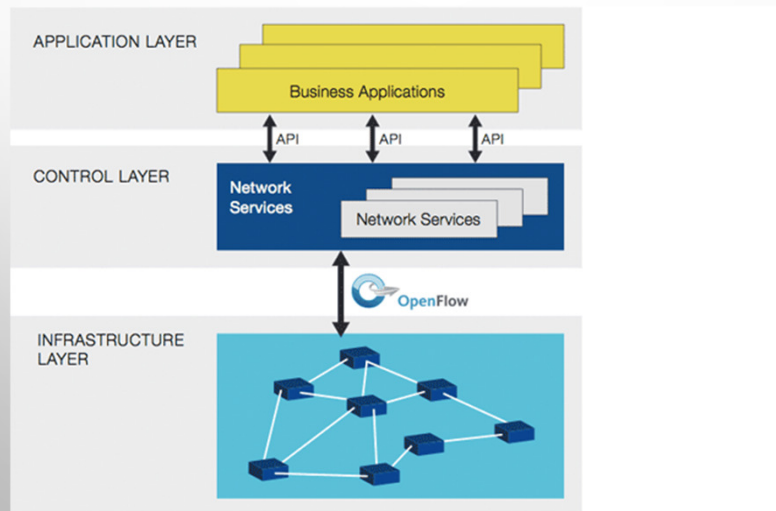
- DIRECTLY PROGRAMMABLE: DECOUPLES NETWORK CONTROL AND FORWARDING SERVICES
- AGILE: LETS ADMINISTRATORS DYNAMICALLY ADJUST NETWORK WIDE TRAFFIC FLOW TO MEET CHANGING NEEDS
- CENTRALLY MANAGED: CENTRALIZED NETWORK INTELLIGENCE, PROGRAMMABLE SDN CONTROLLERS AND MAINTAINS GLOBAL VIEW OF THE NETWORK
- PROGRAMMATICALLY CONFIGURED: NETWORK MANAGERS CAN CONFIGURE, MANAGE, SECURE, AND OPTIMIZE NETWORK RESOURCES VIA DYNAMIC, AUTOMATED SDN PROGRAMS
- OPEN STANDARDS-BASED AND VENDOR-NEUTRAL: IMPLEMENTED THROUGH OPEN STANDARDS AND NOT VENDOR SPECIFIC. HENCE CONTROL IS SIMPLIFIED

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SDN ARCHITECTURE (CONTD...)



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COMPUTING TRENDS: DRIVERS OF NETWORK CHANGE

- CHANGING TRAFFIC PATTERNS
- THE "CONSUMERISATION OF IT"
- THE RISE OF CLOUD SERVICES
- "BIG DATA" MEANS MORE BANDWIDTH

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CHALLENGES FACED IN CURRENT NETWORKS

- COMPLEXITIES THAT LEADS TO STASIS
 - ADDING OR MOVING DEVICES
 - IMPLEMENTING NETWORK POLICIES
 - MANUAL ENDEAVORS LEADS TO NETWORK SERVICE DISRUPTIONS
- INABILITY TO SCALE
- VENDOR DEPENDENCE

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WIRELESS (COMMUNICATION) TECHNOLOGIES

- CELLULAR M2M (AS DISCUSSED EARLIER)
- WIMAX
- WIFI
- BLUETOOTH
- ZIGBEE
- INFRARED

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WIRELESS TECHNOLOGIES (CONTD...)

- WIMAX
 - WORLDWIDE INTEROPERABILITY FOR MICROWAVE ACCESS
 - RADIO WAVES
 - IEEE 802.16 STANDARDS
 - INITIALLY PROVIDED 30 – 40 MEGABIT PER SECOND DATA ACCESS RATE THEN UPDATED TO 1 GIGABITS PER SECOND
 - 4G COMPATIBILITY
 - CAN BE USED FOR
 - TELECOMMUNICATIONS (VOIP)
 - IP TV
 - SMART GRIDS AND METERING

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WIRELESS TECHNOLOGIES (CONTD...)

- WIFI
 - VERY MUCH FAMILIAR TECHNOLOGY FOR ALL, COMMONLY USED TERMINOLOGY, NOT MUCH EXPLANATION REQUIRED
 - RADIO WAVES
 - GENERAL MISCONCEPTION THAT WIFI IS A SHORT FORM FOR WIRELESS FIDELITY. BUT NOT SO
 - WIFI IS A TECHNOLOGY THAT ALLOWS ELECTRONIC DEVICES TO CONNECT TO A WLAN THROUGH ACCESS POINTS (HOT SPOTS)
 - IEEE 802.11X STANDARDS
 - 20 METERS (66 FEET) RANGE AT INDOORS AND MORE AT OUTDOORS
 - INITIALLY PROVIDED 2 MEGABIT PER SECOND DATA ACCESS RATE THEN UPDATED TO 11 MEGABITS PER SECOND

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WIRELESS TECHNOLOGIES (CONTD...)

- BLUETOOTH

- ANOTHER WIRELESS TECHNOLOGY BUT SHORT-RANGE BUILDING PERSONAL AREA NETWORKS (PAN)
- UHF RADIO WAVES
- INVENTED BY ERICSSON IN 1994
- IEEE 802.15.1 STANDARD. BUT IEEE NO MORE MAINTAINS THE STANDARD
- MORE SUITABLE FOR AD HOC NETWORK COMMUNICATIONS
- AVAILABLE IN DIFFERENT VERSIONS 3.0, 4.0 AND 5.0

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WIRELESS TECHNOLOGIES (CONTD...)

- ZIGBEE

- INTENDED FOR WIRELESS PERSONAL AREA NETWORKS (WPAN)
- CAN TRANSMIT DATA OVER LONG DISTANCES THROUGH MESH NETWORKS
 - MESH NETWORK IS A NETWORK TOPOLOGY IN WHICH EVERY NODE RELAYS DATA BY EITHER FLOODING OR ROUTING TECHNIQUE
 - MESH NETWORK IS ANOTHER TYPE OF AD HOC NETWORK (MANET)
- LOW POWER, LOW COST, WIRELESS MESH NETWORK STANDARD TARGETS DEVELOPMENT OF LONG BATTERY LIFE DEVICES
- LOW LATENCY REDUCES AVERAGE CURRENT
- IEEE 802.15.4 STANDARD
- BEST SUITED FOR INTERNET OF THINGS

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WIRELESS TECHNOLOGIES (CONTD...)

- INFRARED
 - INVISIBLE RADIANT ENERGY, ELECTROMAGNETIC RADIATION
 - VERY SHORT RANGE
 - TV REMOTES

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MOBILE COMPUTING - DEVICES

- LAPTOP
- PALMTOP
- PDA/ORGANISER
- MOBILE PHONE
- SATELLITE PHONE
- PAGER
- WEARABLE
- ETC

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MOBILE COMPUTING – APPLICATIONS

- VEHICLES
- NOMADIC USER
- SMART MOBILE PHONE
- INVISIBLE COMPUTING
- WEARABLE COMPUTING
- TAXI/POLICE/FIRE SQUAD
- DISASTER RELIEF/DISASTER ALARM
- MILITARY/SECURITY

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LIMITATIONS TO MOBILE COMPUTING

- RESOURCE CONSTRAINTS – POWER (BATTERY)
- QUALITY OF SERVICE
- BANDWIDTH
- DYNAMIC CHANGES IN COMMUNICATION ENVIRONMENT – SCALABILITY
- INTEROPERABILITY
- SECURITY

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THANK YOU

END OF
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